



Professional Data Analysis & Intelligent Reporting

Professional Data Analysis Report

Comprehensive Analysis & Insights

Report Generated:	December 23, 2025
Platform:	Insightify Professional Analytics
Engine Version:	3.0 (Enhanced AI Analytics)
Report Status:	Confidential

Generated by Insightify™ - AI-Powered Data Analysis Platform
Transforming Data into Actionable Insights

■ Executive Summary

Report Overview:

This comprehensive data analysis report presents key insights and actionable metrics from the provided dataset. Our analysis covers **50,000** records across **11** columns, utilizing advanced statistical methods and visualization techniques.

Dataset Composition:

- Numeric Variables: **5** columns
- Categorical Variables: **6** columns
- Temporal Variables: **0** columns
- Data Completeness: **0** missing values detected
- Data Uniqueness: **0** duplicate rows identified
- Memory Footprint: **19.85 MB**

■ Key Insights & Findings

Our analysis has uncovered the following significant patterns:

1. Transmission is highly concentrated: 'Manual' represents 50.3% of data
2. Sales_Classification is highly concentrated: 'Low' represents 69.5% of data

■ Numeric Analysis

Column	Mean	Median	Std Dev	Min	Max	IQR
Year	2017.02	2017.00	4.32	2010.00	2024.00	8.00
Engine_Size_L	3.25	3.20	1.01	1.50	5.00	1.70
Mileage_KM	100307.20	100388.50	57941.51	3.00	199996.00	100452.25
Price_USD	75034.60	75011.50	25998.25	30000.00	119998.00	45193.50
Sales_Volume	5067.51	5087.00	2856.77	100.00	9999.00	4949.25

■ Categorical Analysis

Category: Model

Unique Values: 11 | Most Frequent: '7 Series' (9.3% of data)

Value	Count	Percentage
7 Series	4,666	10.2%
i3	4,618	10.1%
i8	4,606	10.1%
3 Series	4,595	10.1%
5 Series	4,592	10.1%
X1	4,570	10.0%
X3	4,497	9.9%
X5	4,487	9.8%
M5	4,478	9.8%
X6	4,478	9.8%

Category: Region

Unique Values: 6 | Most Frequent: 'Asia' (16.9% of data)

Value	Count	Percentage
Asia	8,454	16.9%
Middle East	8,373	16.7%
North America	8,335	16.7%
Europe	8,334	16.7%
Africa	8,253	16.5%
South America	8,251	16.5%

Category: Color

Unique Values: 6 | Most Frequent: 'Red' (16.9% of data)

Value	Count	Percentage
Red	8,463	16.9%
Silver	8,350	16.7%
Grey	8,348	16.7%
White	8,304	16.6%
Black	8,273	16.5%
Blue	8,262	16.5%

Category: Fuel_Type

Unique Values: 4 | Most Frequent: 'Hybrid' (25.4% of data)

Value	Count	Percentage
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Hybrid	12,716	25.4%
Petrol	12,550	25.1%
Electric	12,471	24.9%
Diesel	12,263	24.5%

Category: Transmission

Unique Values: 2 | Most Frequent: 'Manual' (50.3% of data)

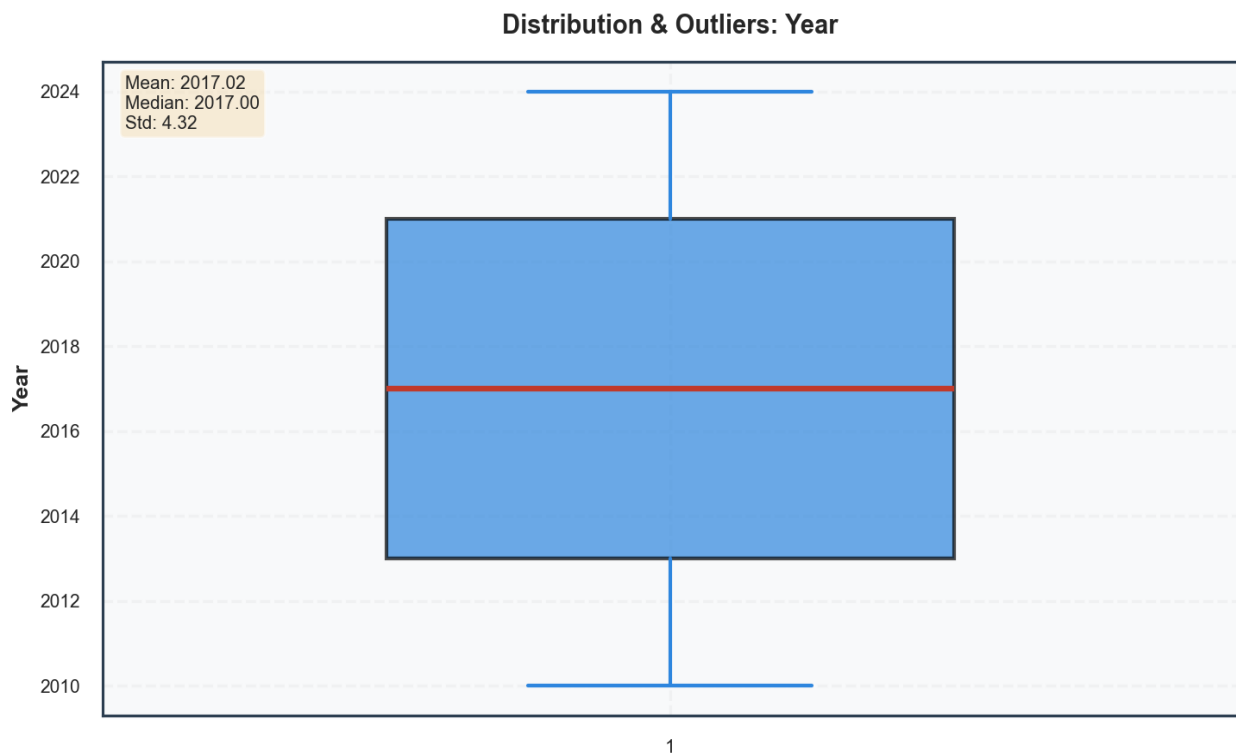
Value	Count	Percentage
Manual	25,154	50.3%
Automatic	24,846	49.7%

■ Correlation Analysis

No significant correlations detected in the dataset.

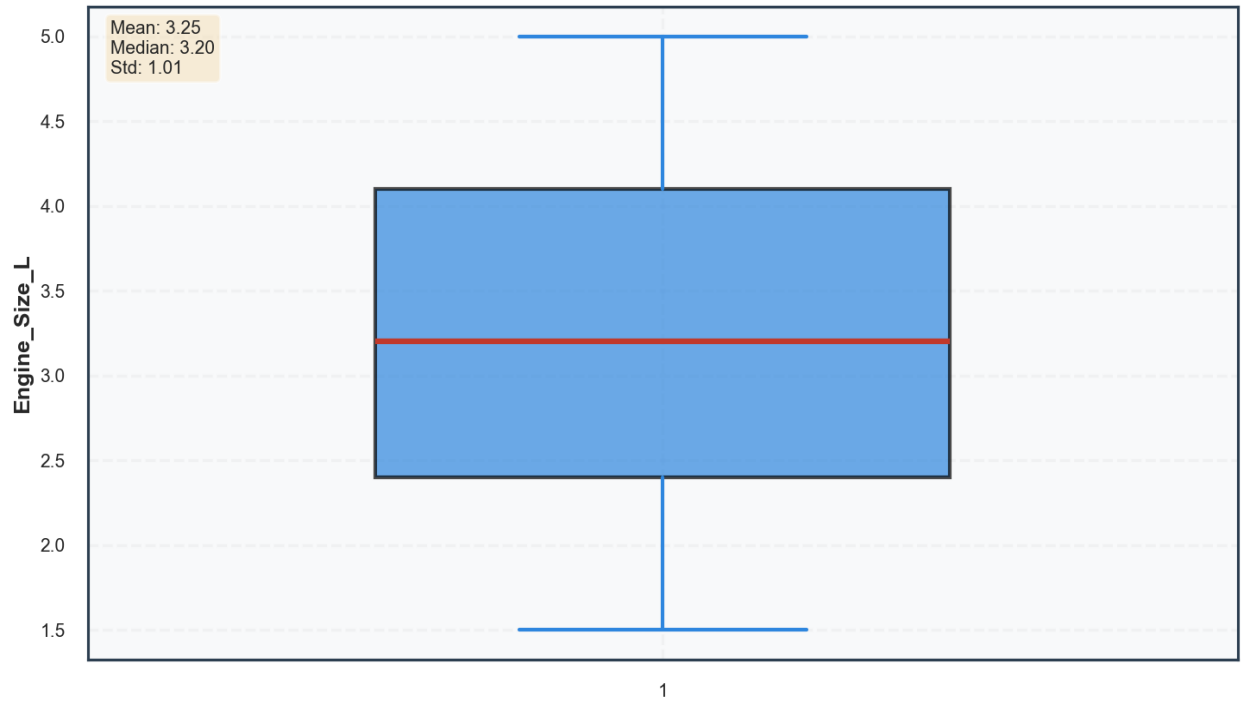
■ Data Visualizations

Boxplot Year

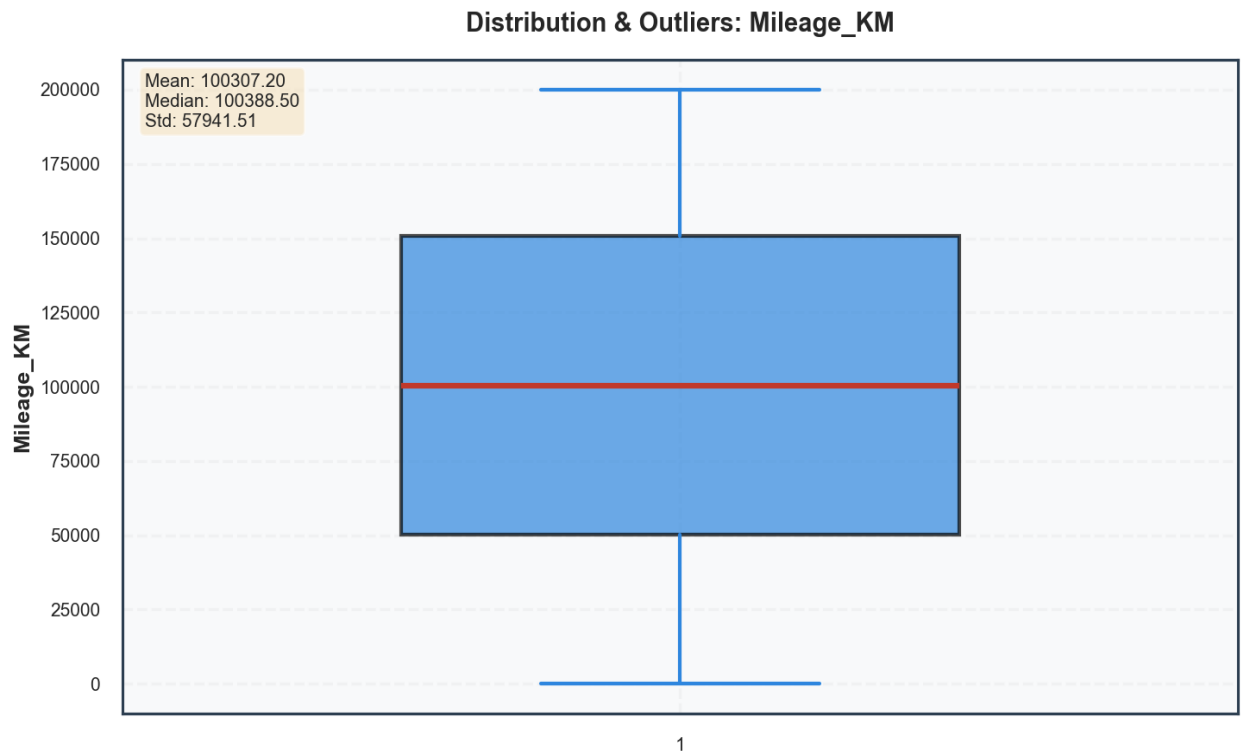


Boxplot Engine Size L

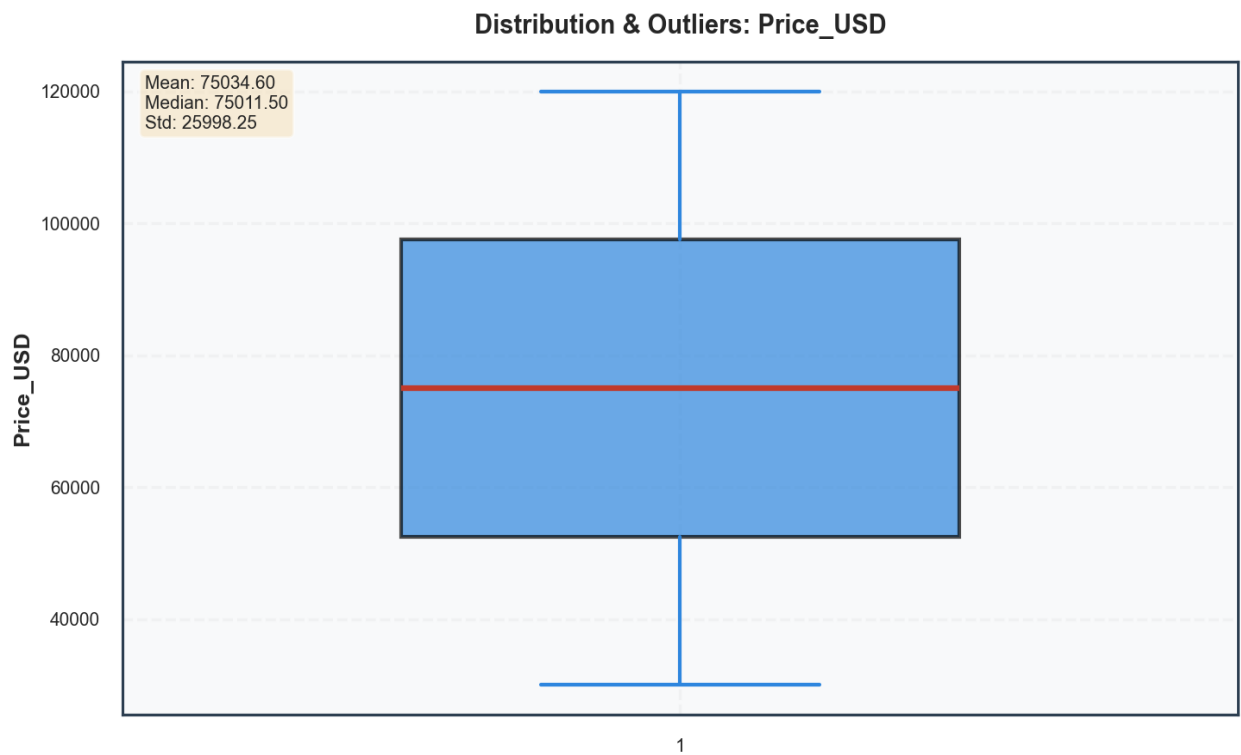
Distribution & Outliers: Engine_Size_L



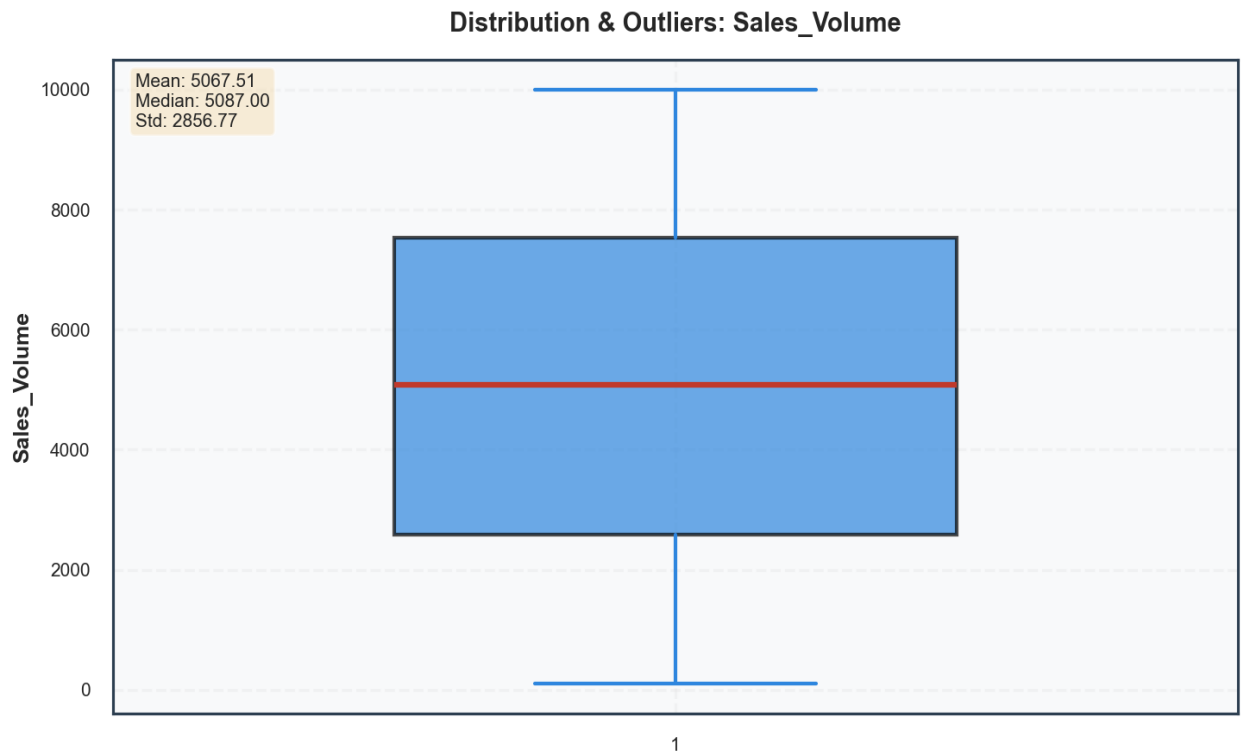
Boxplot Mileage_Km



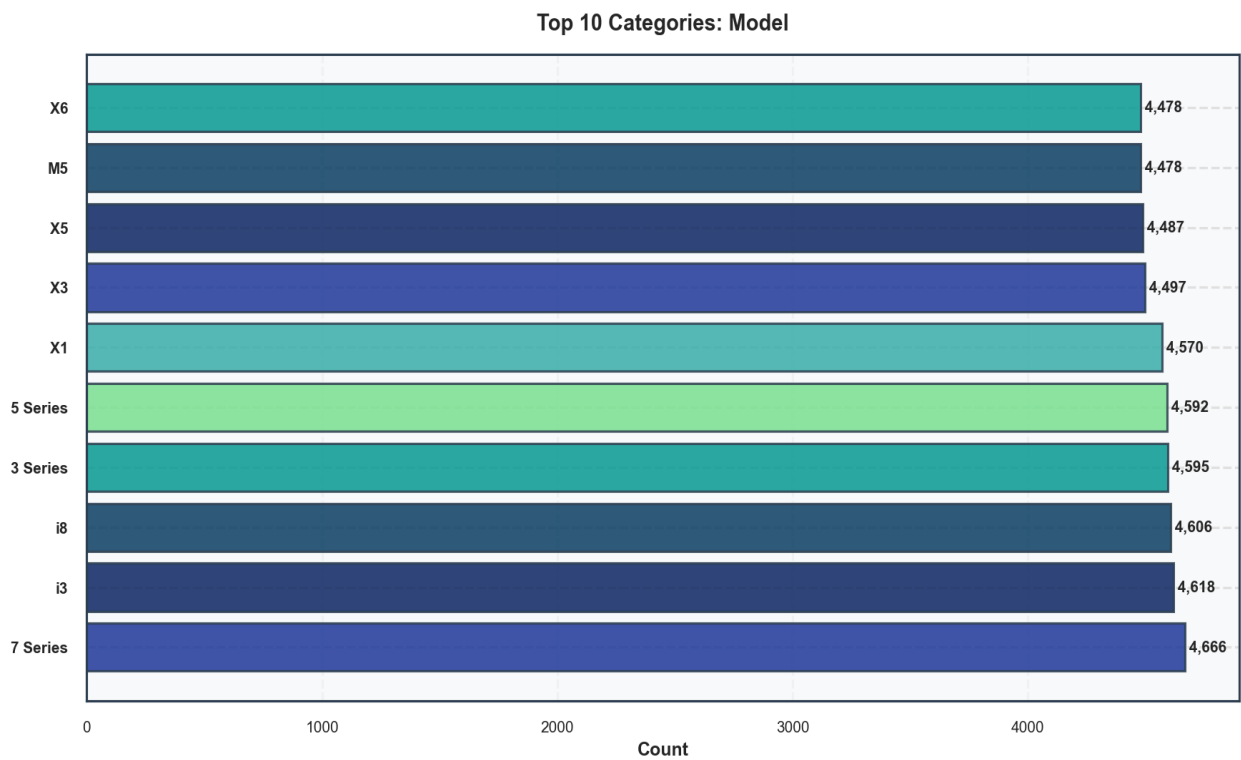
Boxplot Price_Usd



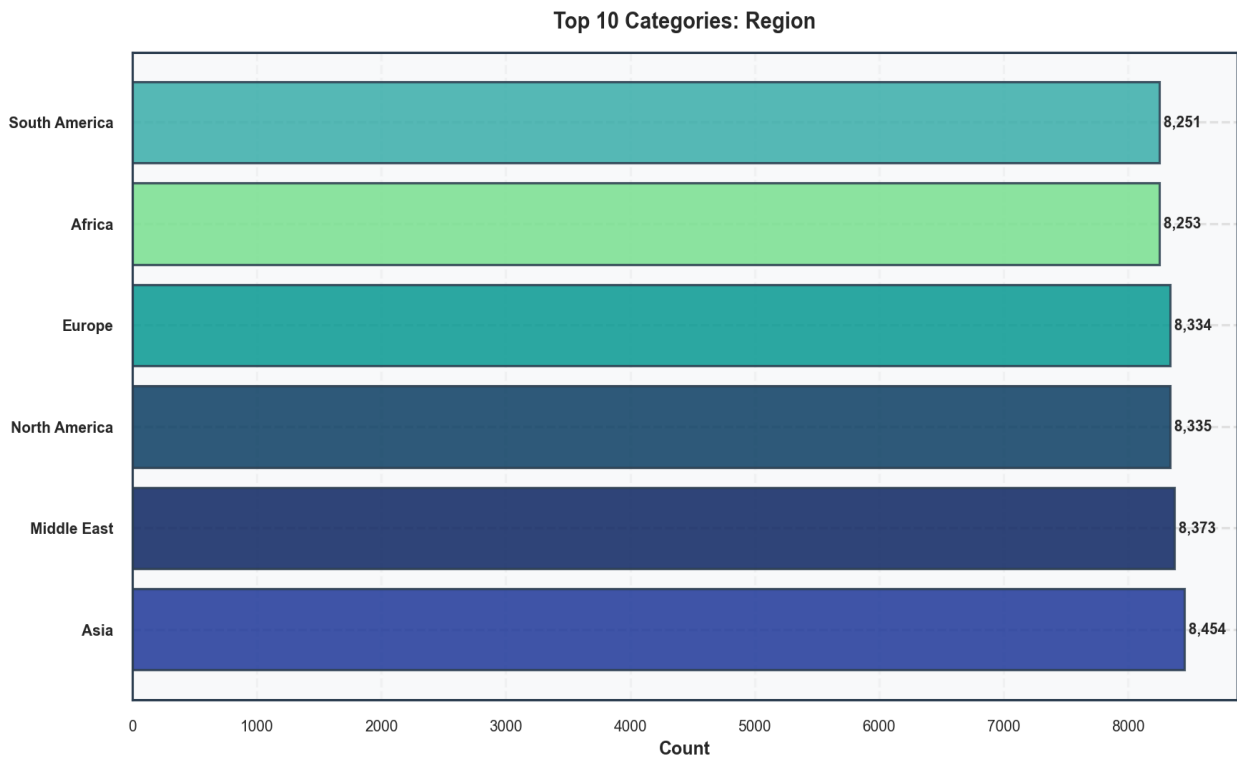
Boxplot Sales Volume



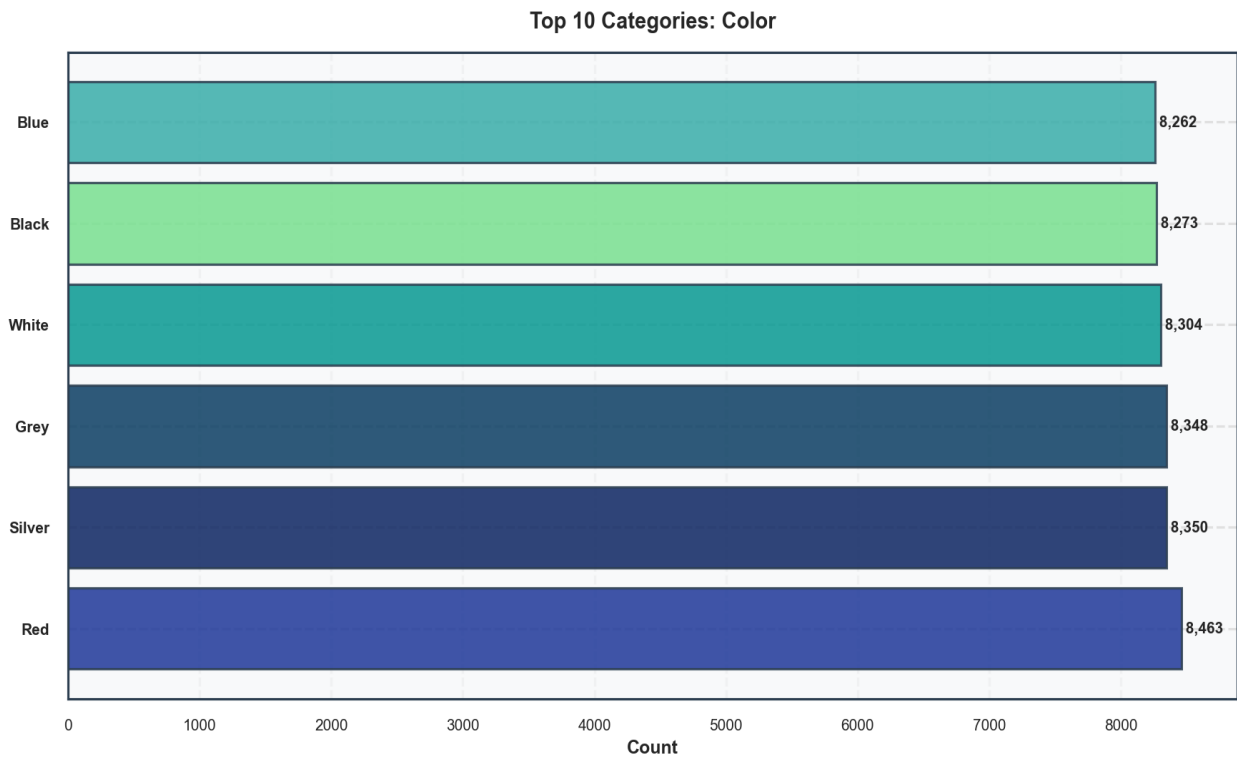
Bar Model



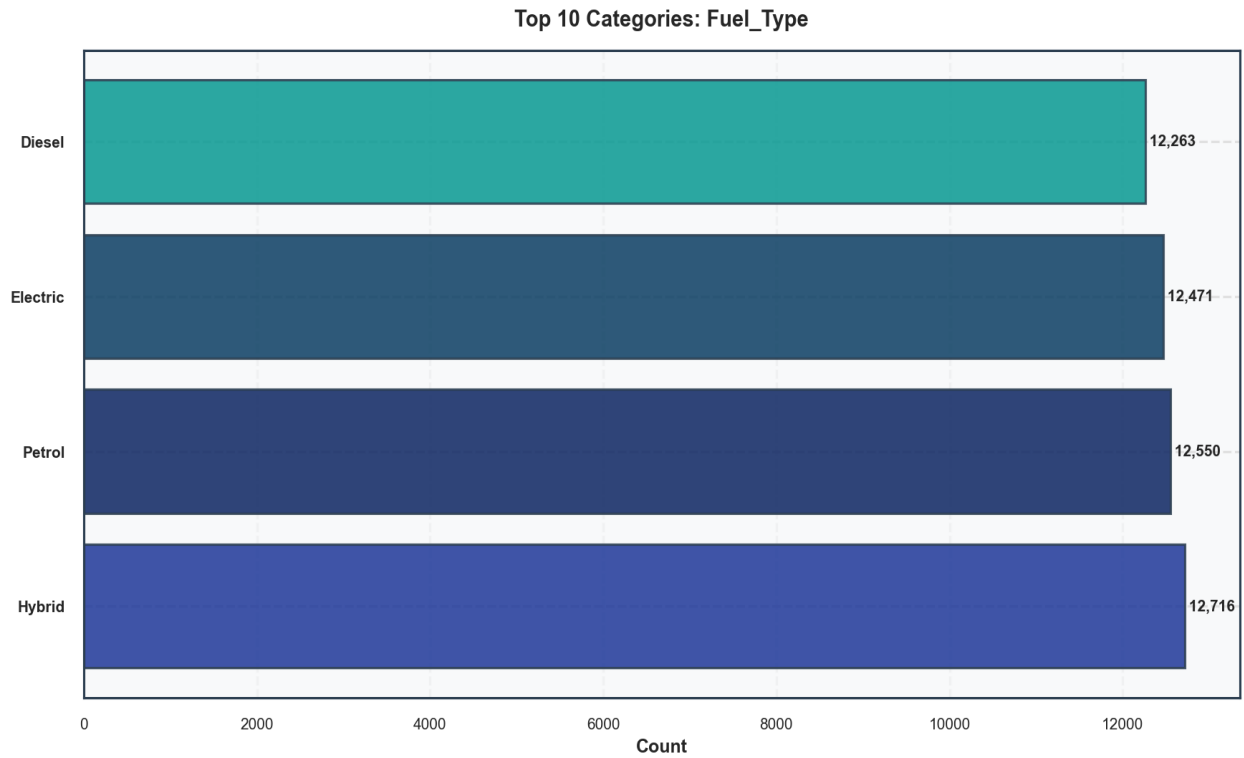
Bar Region



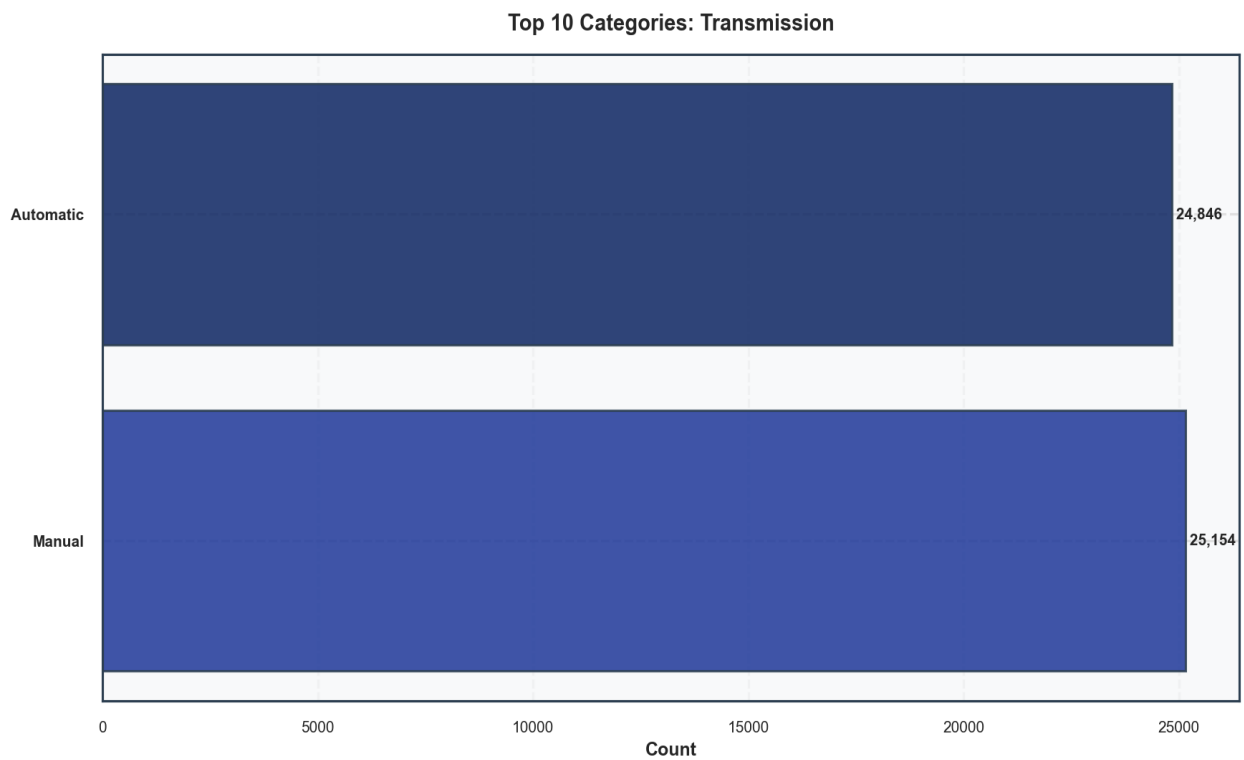
Bar Color



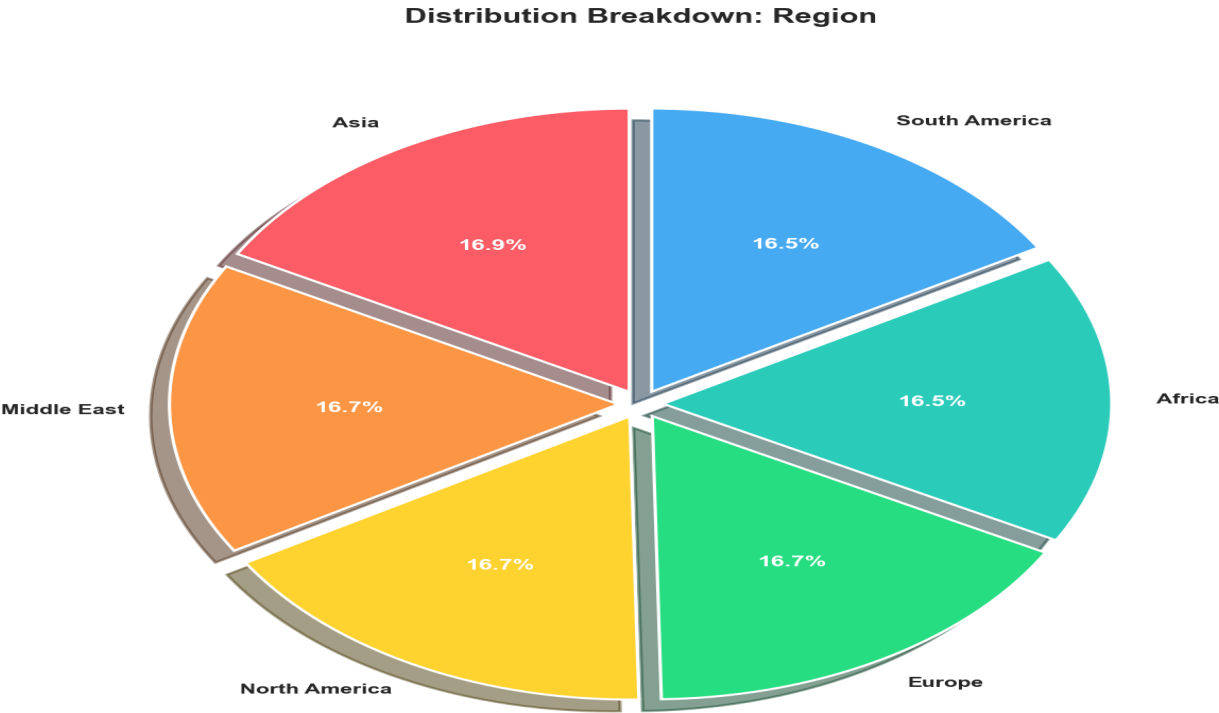
Bar Fuel Type



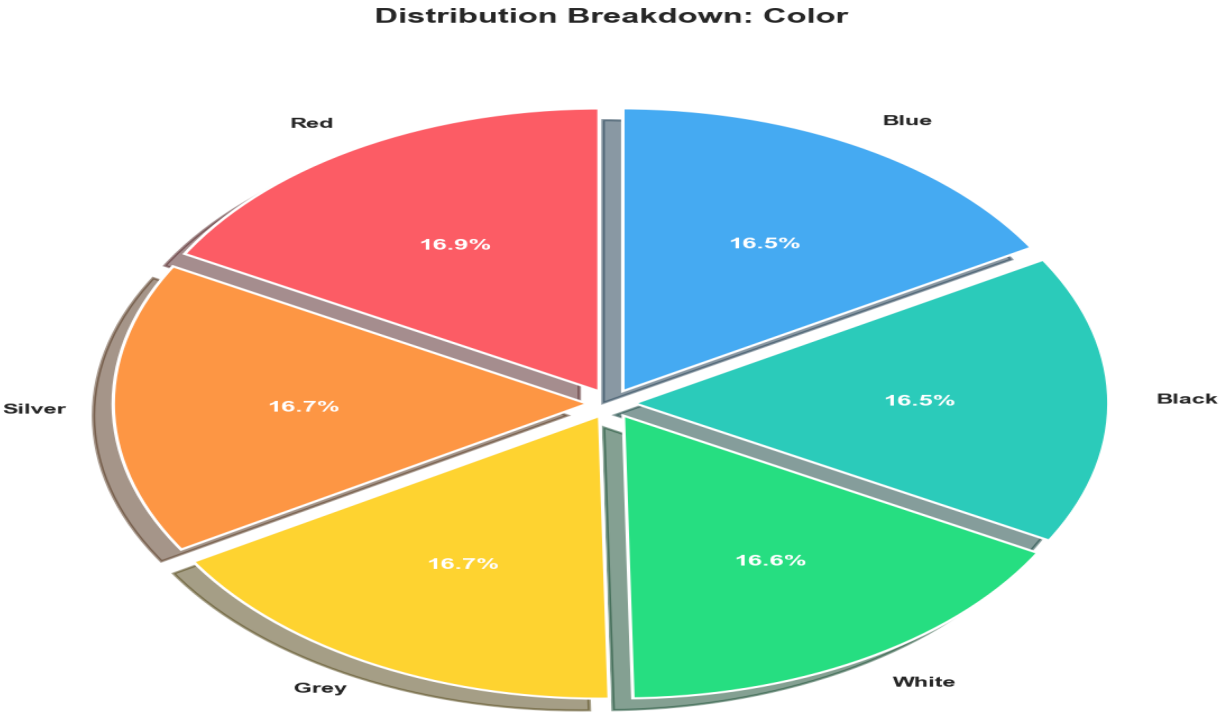
Bar Transmission



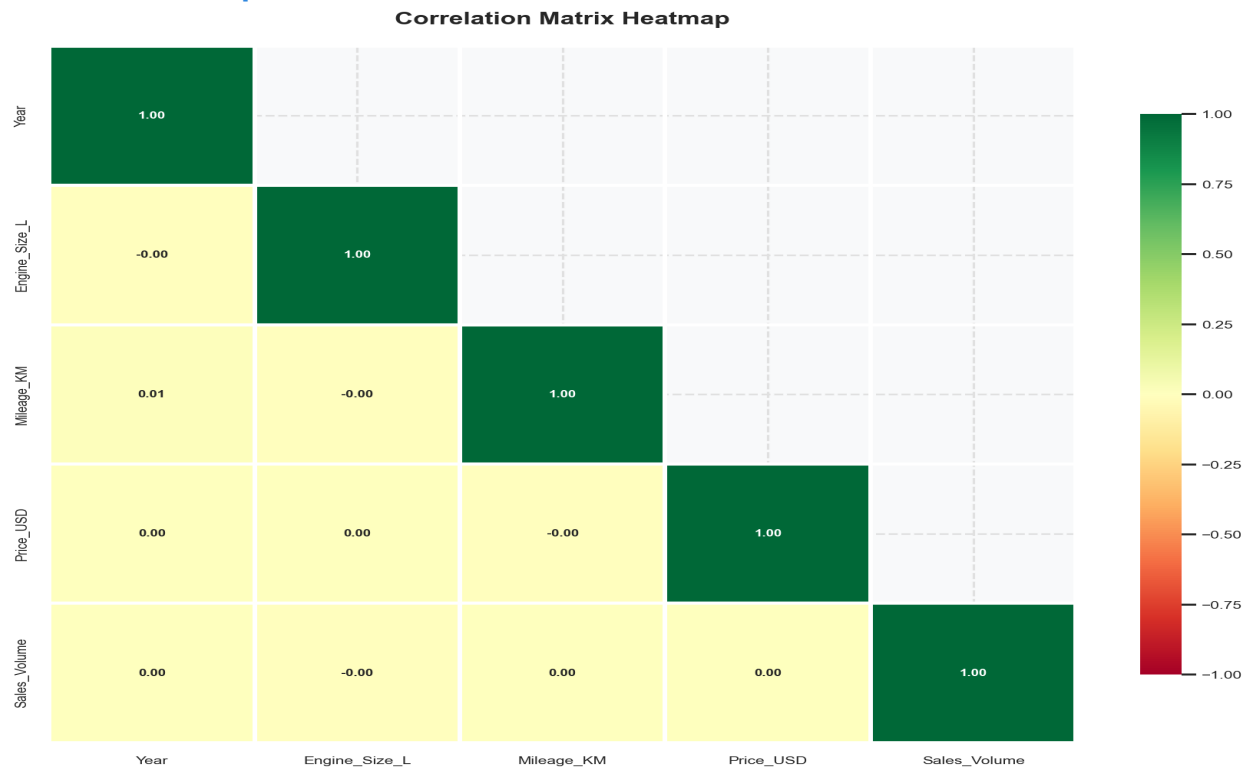
Pie Region



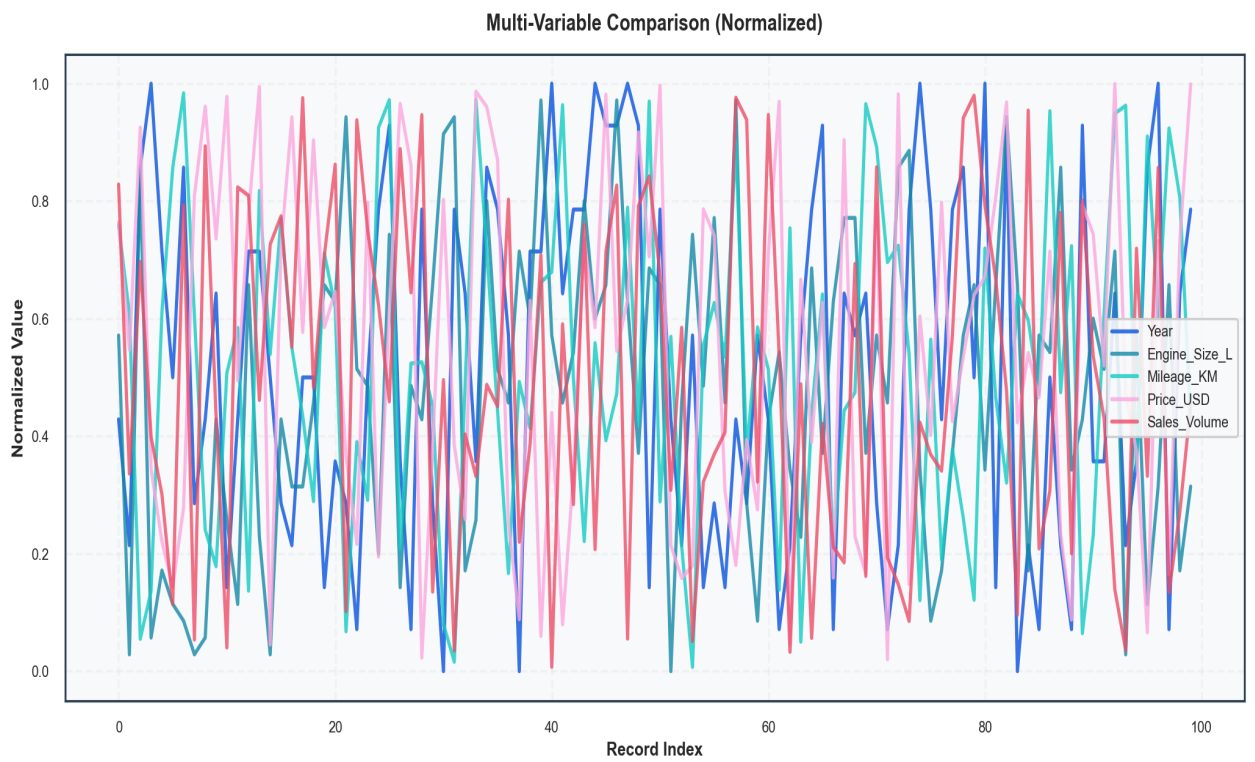
Pie Color



Correlation Heatmap



Multiline Comparison



■ Conclusions & Recommendations

Key Findings:

1. Data Quality Assessment:

The dataset has undergone comprehensive quality checks. Review the Executive Summary for critical insights regarding data completeness, duplicate records, and overall integrity. Address any missing values or duplicates identified before proceeding with predictive modeling.

2. Statistical Distribution Analysis:

Our numeric analysis reveals the central tendencies, variability, and distribution patterns across all quantitative variables. Box plots highlight potential outliers that may require further investigation or treatment depending on your analytical objectives.

3. Correlation & Relationship Discovery:

The correlation analysis has identified statistically significant relationships between variables. Strong positive or negative correlations ($|r| > 0.7$) indicate potential predictive relationships that could be leveraged in machine learning models or business intelligence applications.

Actionable Recommendations:

✓ **Investigate High-Correlation Pairs:** Variables with strong correlations warrant deeper analysis to determine causality versus mere association. Consider domain expertise when interpreting these relationships.

✓ **Address Data Quality Issues:** Implement data cleaning protocols for missing values and duplicates. Consider imputation strategies or removal based on the nature and extent of missing data.

✓ **Leverage Visualizations for Stakeholder Communication:** The generated charts provide clear, professional representations of your data suitable for executive presentations and decision-making forums.

✓ **Monitor Categorical Distributions:** Highly concentrated categories or imbalanced classes may require special handling in predictive models (e.g., SMOTE for oversampling, stratified sampling).

✓ **Temporal Pattern Exploration:** If time-series trends are present, consider seasonal decomposition and forecasting techniques to predict future patterns.

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